LAN and WAN ports of a router (Image courtesy: wikimedia commons)

20MHz, 40MHz, 80MHz and 160MHz widths. Additionally, the 802.11ac (also known as 11ac) standard taps into the 5GHz bandwidth, avoiding clustered or jammed network. The cost, however, is also more.

The most popular category till now, keeping in mind the cost, is the 802.11n standard (also denoted as 11n), which can offer data throughput of up to 600Mbps (considering four spatial streams, each supporting 150Mbps). It supports wireless range of up to 122 metres (400 feet) and two channel widths of 20MHz and 40MHz, and functions in the 2.4GHz bandwidth. While 11ac is the recommended Wi-Fi standard, the choice largely depends on your budget.

More antennae and MU-MIMO for stronger network

The number of antennae on a router indicates the spatial streams (or channels) of data transmission. Each antenna in a router comprises a single transmitter and a single receiver end, and has a notation of its own: 1x1 implies a single antenna with one transmitter and one receiver, and is called a single-input single-output (SISO) system; 2x2 implies two antennae, each with one transmitter and one receiver, and is called a multiple-input multiple-output (MIMO) system.

Routers with a single antenna can transmit data through a single spatial stream—correlating to the 2.4GHz bandwidth. As discussed earlier, in that case, most electronic devices have a chance of interference, causing the network to slow down. 2x2 routers can tap into two spatial streams. In 11n standard routers, both antennae support 2.4GHz bandwidth, attaining a throughput of up to 300Mbps (considering each stream supports 150Mbps data at 2.4GHz).

In comparison, in 11ac standard 2x2 routers, one stream taps into the 5GHz bandwidth, while the other works at 2.4GHz, greatly reducing the cluttering and increasing the data transmission rate from 600Mbps up to 1Gbps.

Another major advantage of 11ac 2x2 or higher-standard routers is that these support multi-user MIMO (MU-MIMO). To explain further, any router standard below ac, including 11n, can transmit data to only one device (single user) at a specific time interval, causing other connected devices to wait for the data, irrespective of the antenna. This results in a comparatively slow speed when multiple devices connect to the same router.

On the other hand, 11ac routers can interact with multiple devices (generally four devices) simultaneously at a given instance, exponentially increasing the network experience. In simpler words, routers with more advanced IEEE standard ratings perform better.

Bonus features

You may consider a few more features depending on your business requirements. For instance, check whether the router supports IPv6 for the new-generation networking. To access your business network from a distant location, VPN compatibility is required. Protocol examples include PPTP, L2TP, IPSec and so on. A router that hosts multiple service set identifiers (SSIDs) can be useful for separate guest access to the business network.

A glimpse into the market

For typical needs of small offices or home offices, a single router can cost anywhere between ₹ 1500 and ₹ 4000. Some of the prominent brands include D-Link, TP-Link, Asus, Netgear, Linksys, Cisco, Binatone and Buffalo. The Internet is the backbone of businesses today. Therefore finding the right router for the purpose has no alternatives. EFY